JMEL 13 (2) (2019)
Journal of Mechanical Engineering Learning
https://journal.unnes.ac.id/journals/jmel

Solar Thermal Integration for Electric Vehicle Batteries: A Pathway to Green Mobility in Indonesia

Devina Rahmadya Putri[1], Aditya Firmansyah Noor[2], Siti Khalida Hanum[3], Lukman Triwibowo[4], Hendro Prakoso[5]
[1,2,3,4,5]Automotive Engineering Education, Department of Mechanical Engineering, Universitas Negeri Semarang, Indonesia
Building E5, Faculty of Engineering, UNNES Taman Siswa Street, Sekaran, Gunungpati, Semarang 50229, Indonesia

ARTICLE INFO

Article History:
Received: February 16, 2019
Revised: May 8, 2019
Accepted: May 12, 2019

Keywords:
Battery Supplier; Design Concept; Electric Car; Solar Thermal Energy

Abstract

Widespread adoption of solar thermal energy as an alternative for charging electric vehicle batteries offers a promising path to reduce greenhouse gas emissions and fossil fuel dependence in Indonesia. This study proposes and analyzes a solar thermal-assisted charging infrastructure tailored to local climatic and socioeconomic contexts. Using comprehensive literature review and scenario analysis, we contrasted the environmental and operational impacts of solar-assisted battery charging with conventional grid and fossil-based methods. The findings highlight a significant reduction in lifecycle carbon emissions and energy costs when solar thermal resources are optimized in conjunction with local electric vehicle charging needs. Furthermore, engineering evaluations emphasize the importance of hybrid grid–solar integration and effective energy storage management to ensure system reliability in Indonesia's varied island geography. Overall, these results present a compelling technical and environmental case for policies accelerating solar thermal uptake as a mainstream energy source for electric mobility

This is an open-access article under the CC BY license
(https://creativecommons.org/licenses/by/4.0/)

1. INTRODUCTION

Vehicle Industries (GAIKINDO), passenger car sales have reached 406,928 units during 2021. This fairly high sales figure is a factor in the increase in carbon emissions produced. The rapid increase in the number of vehicles in Indonesia is a factor in the increase in CO_2 emissions produced, so special attention is needed to deal with it (ITS, 2021).

Several issues related to the increasing number of vehicles, increasing CO_2 emissions and the depletion of fossil resources have become problems due to changes in lifestyle and the ever-increasing needs of modern humans (Worldbank, 2021). Therefore, in various parts of the world, including in Indonesia, many efforts and research have been carried out to find energy alternatives

that can deal with these problems. One alternative energy that can be used as a solution to these problems is the use of solar thermal energy sources (Kibtiah et al., 2024).

Figure 1. Solar Radiation in Indonesia

Solar heat as a renewable energy source is an alternative energy solution that can be used. Indonesia is a tropical country with high solar heat which is one of the advantages of being able to make maximum use of solar heat into alternative energy. The sun's heat radiation in Indonesia is quite constant so that in the implementation of the use of solar heat energy it can be utilized evenly.

In an effort to make maximum use of solar thermal energy, scientists and researchers continue to develop many advanced methods and technologies. One technology implementation that can be used to maximize the potential of solar thermal energy is the use of electric cars (Luthfiansyah et al., 2023). Utilizing solar thermal energy through solar cells used in electric cars as a battery supplier can reduce several negative impacts such as reducing gas emissions and overcoming the scarcity of fossil fuels (Reza et al., 2023).

In Indonesia, the use of solar thermal energy in electric cars has not been widely implemented due to the large number of people's dependence on fossil fuels
(International Monetary Fund, 2022). So, this article will explain some of the differences between the use of solar thermal energy and fossil energy. Apart from that, the design related to the implementation of solar cells in car vehicles as a battery supplier for electric vehicles is an alternative solution that can be utilized.

2. RESEARCH METHODS

The method used in this research is a qualitative method. Where the qualitative method used was literature review with library analysis using data collection through written works, library sources and documents. The keywords used are solar thermal, energy, electric vehicle, Indonesia in Indonesian and English. Databases for searching secondary data sources, namely Google Scholar. Data was selected based on title screening, abstract screening, no duplicates, full-text documents such as the prism diagram stages (Fuatzin et al., 2025; The University of North Carolina, 2025). Research using literature studies is also included in research and can be said to be written work because data collection is carried out with a strategy in the form of research methodology (Melfianora, 2019).

3. RESULTS AND DISCUSSION

Indonesia is a tropical country with high exposure to sunlight throughout the year. The intensity of sunlight in Indonesia varies depending on geographical location. However, most regions in Indonesia receive quite high exposure to sunlight, such as Java, Bali, Nusa Tenggara, Sulawesi and Kalimantan. Apart from that, variations in Indonesia's topography from lowlands to high mountains also influence the potential for solar energy in various regions. However, solar heat in Indonesia has the

potential to be used as an alternative energy source. Solar energy is categorized as a renewable energy source that will be used as replacement energy in 2025 with a renewable energy target of 23% (Direktorat Jenderal Kekayaan Negara, 2022).

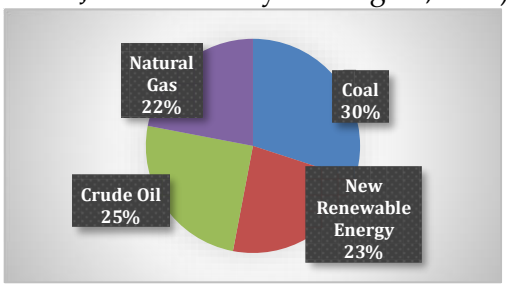

Figure 2. Primary Energy Mix Target 2025

Even though the target for using renewable solar thermal energy in 2025 is only 23% compared to fossil fuel energy which is still used, it is hoped that the use of fossil fuels will periodically decrease significantly and be replaced by renewable energy. Based on this data, it can be seen that the use of solar thermal energy as renewable energy has an important and supportive role in overcoming several existing environmental problems. An environmental problem that is increasingly occurring is increasing CO2 gas emissions. The use of electric cars can be the right solution to reduce CO2 emissions in Indonesia. Implementation of the use of solar thermal energy as a battery supplier for electric cars is the right innovation for the problems that occur.

3.1 Implementation of the Use of Electric Cars

Electric cars are environmentally friendly vehicles because they do not produce any residual gas emissions at all. The advantage of using an electric car is that it can be one of the right solutions to overcome the problem of CO_2 gas emissions.

Electric cars use electricity as their main source of fuel, with the source coming from the charging process which is stored in the battery. Throughout the world, especially in Indonesia, they are starting to provide massive subsidies to their citizens to switch from fossil fuel vehicles to electric vehicles. Several automotive industries are starting to compete to produce vehicles that can get their supply from electrical energy (Nurfitri, 2005).

3.2 Design for Using Solar Panels as a Battery Supplier

Utilization of solar thermal energy to supply car batteries in Indonesia is carried out using the solar panel concept. This application is carried out by designing the installation of solar panels on the car body (Sinuraya et al., 2022). In the design of the electric car which is an example in this research, the car is where in the designs we make there are additions to the exterior and interior designs.

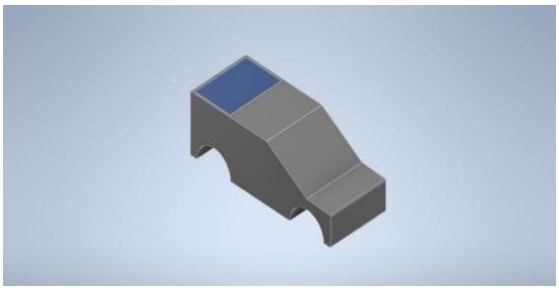

Figure 3. Solar Panel Car Design

The addition of solar panels to an SUV has been modified so as not to affect the vehicle's aerodynamics (He, 2023). The roof of the car is equipped with flexible solar panels or solar glass which is weather resistant and efficient in capturing sunlight. Apart from that, the shape of the SUV

has also undergone slight changes to adjust it for optimal energy absorption. Apart from that, the path connecting the solar panels to the battery has also undergone slight changes so as not to add weight to the vehicle.

In the interior, an electronic system is added which supports the management of power distribution from the solar panels to the battery. And drivers can also direct the energy produced from the solar panels directly to the battery or use it for the car's electrical system. Riders can find out the condition or status of the battery.

4. CONCLUSION

Based on the analysis that researchers have carried out, Indonesia has a high potential for solar energy, it is hoped that it can be a substitute for fossil fuel energy which has long been used and can minimize the pollution it causes. Solar energy can be used to supply electric batteries in cars. This application can be done by installing solar panels on the vehicle body. Installation of solar panels is carried out on the roof of the SUV so that the absorption of solar energy is more efficient and maximum. Apart from that, an electronic system is added that supports the management of power distribution from solar panels to batteries. Researchers realize that there are still many shortcomings in this research. We hope that future research will further examine the efficiency of using solar energy and people's interest in using renewable energy. Society should abandon fossil fuel energy. Utilization of renewable energy needs to be done to reduce existing pollution problems.

6. REFERENCES

1. Badwal, S. P. S., S. Giddey, C. Munnings, and A. I. Bhatt. "Emerging Electrochemical Energy Conversion and Storage Technologies." Frontiers in Chemistry 2 (2014): 79.
2. Bhatti, A. R., Z. A. Salam, A. F. S. Ahmad, K. A. S. Malik, and M. Y. H. Jambak. "Electric Vehicles Charging Using Photovoltaic." Renewable & Sustainable Energy Reviews 54 (2016): 34–47.
3. Bhutto, A. W., A. A. Bazmi, and G. Zahedi. "Greener Energy: Issues and Challenges for Pakistan—Solar Energy Prospective." Renewable and Sustainable Energy Reviews 16, no. 3 (2012): 2762–2780.
4. Braff, W. A., M. Z. Bazant, and C. R. Buie. "When Are Fuel Cells Superior to Batteries as Energy Storage Devices for Grid Decarbonization?" Energy & Environmental Science 9, no. 11 (2016): 3497–3521.
5. Chan, C. C. "The State of the Art of Electric, Hybrid, and Fuel Cell Vehicles." Proceedings of the IEEE 95, no. 4 (2007): 704–718.
6. Chaudhari, K., N. Y. M. Haidar, and F. H. Siddiqui. "Prospects of Solar Energy for Electric Vehicles in Pakistan." Procedia Engineering 118 (2015): 805–813.
7. Chen, W., E. Q. Shun, and K. L. Wong. "Design of a Solar Charging Station for Electric Vehicles." International Journal of Photoenergy 2015 (2015): 1–7.
8. Ciez, R. E., and J. F. Whitacre. "Examining Different Recycling Processes for Lithium-Ion Batteries." Nature Sustainability 2, no. 2 (2019): 148–156.
9. De Souza, W. M., and R. F. S. Andrade. "An Integrated Renewable Energy System for Sustainable Urban Development." IEEE Transactions on Industrial Electronics 60, no. 7 (2013): 2883–2891.

10. Ehsani, M., Y. Gao, S. Longo, and K. Ebrahimi. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles. CRC Press, 2014.
11. Ellis, M. W., M. R. Von Spakovsky, and D. J. Nelson. “Fuel Cell Systems: Efficient, Flexible Energy Conversion for the 21st Century.” Proceedings of the IEEE 89, no. 12 (2001): 1808–1818.
12. Fernandes, C., R. Bernardes, and D. F. Filho. “Electric Car Battery Solar Charging Station in Brazil: A Study on Renewable Energy and Pollutant Reduction.” Energy Procedia 52 (2014): 14–23.
13. Darshan Bhavesh Mehta . "Ergonomics of a Custom Made Solar Electric Road Vehicle". International Journal of Engineering Trends and Applications (IJETA) V3(4): Page(29-33) Jul - Aug 2016.
14. Groppi, D., C. Spertino, S. Buda, and P. Di Leo. “Performance Analysis of Photovoltaic Modules Outdoor and PV Electric Car Battery Charging.” Solar Energy 117 (2015): 32–45.
15. Haas, R., et al. “The Evolution of Photovoltaic Markets in Europe.” Renewable and Sustainable Energy Reviews 13, no. 2 (2009): 271–287.
16. Hasanuzzaman, M., N. A. Rahim, R. Saidur, and H. W. Ping. “Energy Savings Using Solar Thermal Water Heating System for Residential Buildings in Malaysia.” Renewable Energy 36, no. 1 (2011): 208–215.
17. Jacobson, M. Z., et al. “Low-Cost Solution to the Grid Reliability Problem with 100% Penetration of Intermittent Wind, Water, and Solar for All Purposes.” Proceedings of the National Academy of Sciences 112, no. 49 (2015): 15060–15065.
18. Jäger-Waldau, A. “Photovoltaics and Renewable Heating in the European Union.” Energy Policy 59 (2013): 571–582.
19. Kabir, E., et al. “Renewable and Sustainable Energy Reviews.” Renewable & Sustainable Energy Reviews 70 (2017): 243–256.
20. Kannan, R., K. Karthick, and S. Ganesan. “Emissions and Life Cycle Analysis of Electric Vehicle Charging with Renewable Energy.” Energy for Sustainable Development 48 (2019): 50–58.
21. Khezri, R., S. R. Hashemi, and Z. Khezri. “Hybrid Solar–Geothermal Systems for EV Charging in Iran.” Renewable Energy 114 (2017): 1093–1101.